

Quantum States

Complete Mathematical Framework

Pauli Matrices, Bloch Sphere, Pure & Mixed States,
Density Matrices, and Comprehensive Purity Tests

*"The underlying physical laws necessary for the
mathematical theory of large parts of physics and the
whole of chemistry are thus completely known."*

— Paul Dirac

By: **Machiraju Karthikeya**

COMPREHENSIVE COVERAGE:

- Pauli Matrices: Mathematical Foundation
- Bloch Sphere: Geometric Representation
- Bloch Vectors: Computational Methods
- Pure States: Complete Information
- Mixed States: Statistical Ensembles
- Density Matrices: Universal Formalism
- Purity Tests: The Ultimate Quantum Detector
- Bloch Vector-Based Purity Tests
- Physical Applications and Examples

*"In physics, you don't have to go around making trouble for yourself. Nature does it for
you."*

— Frank Wilczek

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1 Pauli Matrices: The Foundation of Quantum Mechanics

1.1 Mathematical Definition

Definition

The Pauli matrices are the fundamental building blocks of single-qubit quantum mechanics:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (\text{Pauli-X or bit-flip}) \quad (1)$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (\text{Pauli-Y}) \quad (2)$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{Pauli-Z or phase-flip}) \quad (3)$$

Together with the identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, they form a complete basis for all 2×2 Hermitian matrices.

1.2 Fundamental Properties

Theorem

The Pauli matrices satisfy the following essential properties:

1. **Hermitian:** $\sigma_i^\dagger = \sigma_i$ for all $i = x, y, z$
2. **Unitary:** $\sigma_i^\dagger \sigma_i = I$ (they are their own inverses)
3. **Involutory:** $\sigma_i^2 = I$ for all i
4. **Traceless:** $\text{Tr}(\sigma_i) = 0$ for all i
5. **Eigenvalues:** All Pauli matrices have eigenvalues $\lambda = \pm 1$
6. **Determinant:** $\det(\sigma_i) = -1$ for all i

1.3 Commutation and Anticommutation Relations

Key Formula

Commutation Relations:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$$

where ϵ_{ijk} is the Levi-Civita symbol.

Anticommutation Relations:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I$$

where δ_{ij} is the Kronecker delta.

Product Relations:

$$\sigma_i\sigma_j = \delta_{ij}I + i\epsilon_{ijk}\sigma_k$$

1.4 Eigenvalues and Eigenvectors

Example

For σ_x :

$$\text{Eigenvalues: } \lambda = \pm 1 \quad (4)$$

$$\text{Eigenvectors: } |+\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (5)$$

For σ_y :

$$\text{Eigenvalues: } \lambda = \pm 1 \quad (6)$$

$$\text{Eigenvectors: } |+\rangle_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |-\rangle_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (7)$$

For σ_z :

$$\text{Eigenvalues: } \lambda = \pm 1 \quad (8)$$

$$\text{Eigenvectors: } |+\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle, \quad |-\rangle_z = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \quad (9)$$

2 The Bloch Sphere: Geometric Visualization of Quantum States

2.1 Physical and Mathematical Foundation

Definition

The Bloch sphere is a unit sphere in 3D space that provides a geometric representation of all single-qubit quantum states. Every point on or inside the sphere corresponds to a valid quantum state:

- **Surface points:** Pure states ($|\vec{r}| = 1$)
- **Interior points:** Mixed states ($|\vec{r}| < 1$)
- **Center:** Maximally mixed state ($\vec{r} = \vec{0}$)

2.2 Coordinate System

Important Note

Special Points on the Bloch Sphere:

- **North pole** $(0, 0, 1)$: State $|0\rangle$
- **South pole** $(0, 0, -1)$: State $|1\rangle$
- **Positive x-axis** $(1, 0, 0)$: State $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- **Negative x-axis** $(-1, 0, 0)$: State $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- **Positive y-axis** $(0, 1, 0)$: State $|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$
- **Negative y-axis** $(0, -1, 0)$: State $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

2.3 Spherical Parameterization

Key Formula

Any quantum state can be parameterized using spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

The corresponding Bloch vector is:

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

3 Bloch Vectors: The Computational Framework

3.1 Universal Pauli Decomposition

Universal Bloch Representation

Any qubit density matrix can be uniquely written as:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

where:

- $\vec{r} = (r_x, r_y, r_z)$ is the **Bloch vector**
- $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the **Pauli vector**
- $|\vec{r}| \leq 1$ always (fundamental quantum constraint)

3.2 Calculating Bloch Vector Components

Bloch Vector Calculation Methods

Given a density matrix $\rho = \begin{pmatrix} a & b \\ b^* & c \end{pmatrix}$ with $a + c = 1$:

Method 1: Direct Trace Calculation

$$r_x = \text{Tr}(\rho\sigma_x) = 2\text{Re}(b) \quad (10)$$

$$r_y = \text{Tr}(\rho\sigma_y) = -2\text{Im}(b) \quad (11)$$

$$r_z = \text{Tr}(\rho\sigma_z) = a - c \quad (12)$$

Method 2: From Matrix Elements If $b = |b|e^{i\phi}$, then:

$$r_x = 2|b| \cos \phi \quad (13)$$

$$r_y = -2|b| \sin \phi \quad (14)$$

$$r_z = a - c \quad (15)$$

Verification: $|\vec{r}|^2 = r_x^2 + r_y^2 + r_z^2 = 4|b|^2 + (a - c)^2$

4 Pure States: Complete Quantum Information

4.1 Mathematical Definition

Definition

A quantum state is **pure** if it can be described by a single state vector $|\psi\rangle$.

Density Matrix Representation:

$$\rho = |\psi\rangle\langle\psi|$$

Fundamental Characteristics:

- Rank-1 projector: $\rho^2 = \rho$
- Idempotent: $\rho = \rho^2$
- Maximum purity: $\text{Tr}(\rho^2) = 1$
- On Bloch sphere surface: $|\vec{r}| = 1$

4.2 Examples of Pure States

Computational Basis State $|0\rangle$

State vector: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Density matrix:

$$\rho = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Purity check:

$$\rho^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \rho$$

$$\text{Tr}(\rho^2) = 1 \quad \text{CONFIRMED PURE}$$

Bloch vector: $\vec{r} = (0, 0, 1)$ with $|\vec{r}| = 1$

Superposition State $|+\rangle$

State vector: $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Density matrix:

$$\rho = |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Purity verification:

$$\rho^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \rho$$

$$\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1 \quad \text{CONFIRMED PURE}$$

Bloch vector: $\vec{r} = (1, 0, 0)$ with $|\vec{r}| = 1$

5 Mixed States: Statistical Ensembles

5.1 Mathematical Definition

Definition

A quantum state is **mixed** if it represents incomplete knowledge about a quantum system.

Mathematical Representation:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where $p_i \geq 0$, $\sum_i p_i = 1$, and $\{|\psi_i\rangle\}$ are normalized states.

Fundamental Characteristics:

- Non-idempotent: $\rho^2 \neq \rho$
- Reduced purity: $\text{Tr}(\rho^2) < 1$
- Inside Bloch sphere: $|\vec{r}| < 1$
- Rank > 1 (generally)

5.2 Examples of Mixed States

Classical Mixture of Basis States

System preparation:

- Probability 0.7: state $|0\rangle$
- Probability 0.3: state $|1\rangle$

Density matrix:

$$\rho = 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1| = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.3 \end{pmatrix}$$

Purity calculation:

$$\rho^2 = \begin{pmatrix} 0.49 & 0 \\ 0 & 0.09 \end{pmatrix}$$

$$\text{Tr}(\rho^2) = 0.49 + 0.09 = 0.58 < 1 \quad \text{CONFIRMED MIXED}$$

Bloch vector: $\vec{r} = (0, 0, 0.4)$ with $|\vec{r}| = 0.4 < 1$

The Maximally Mixed State

Definition: The state with minimum purity for a given dimension.

For a qubit: $\rho = \frac{1}{2}I = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Purity analysis:

$$\rho^2 = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{4}I$$

$$\text{Tr}(\rho^2) = \frac{1}{4}(1 + 1) = \frac{1}{2}$$

This is the minimum possible purity for any qubit state!

Bloch vector: $\vec{r} = (0, 0, 0)$ (Center of Bloch sphere)

Multiple decompositions possible:

$$\text{Computational basis: } \rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| \quad (16)$$

$$\text{Hadamard basis: } \rho = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| \quad (17)$$

6 Density Matrices: Universal Quantum Formalism

6.1 General Properties

Density Matrix Requirements

Any valid density matrix ρ must satisfy:

1. **Hermitian:** $\rho^\dagger = \rho$ (ensures real eigenvalues)
2. **Positive semi-definite:** $\rho \geq 0$ (ensures non-negative probabilities)
3. **Unit trace:** $\text{Tr}(\rho) = 1$ (probability conservation)
4. **Physical interpretation:** $\rho_{ij} = \langle i|\rho|j\rangle$ represents quantum amplitudes

6.2 Thermal Quantum States

Thermal Equilibrium Analysis

System: Qubit with energy gap $\Delta E = \hbar\omega$

Thermal density matrix:

$$\rho_{\text{thermal}} = \frac{1}{Z} e^{-\beta H}$$

where $\beta = \frac{1}{k_B T}$ and $Z = \text{Tr}(e^{-\beta H})$

For a qubit with $H = \frac{\hbar\omega}{2} \sigma_z$:

$$\rho_{\text{thermal}} = \frac{1}{2} \begin{pmatrix} 1 + \tanh(\beta\hbar\omega/2) & 0 \\ 0 & 1 - \tanh(\beta\hbar\omega/2) \end{pmatrix}$$

Bloch vector: $\vec{r}_{\text{thermal}} = (0, 0, \tanh(\beta\hbar\omega/2))$

Temperature limits:

- **Low temperature** ($T \rightarrow 0$): $\tanh(\beta\hbar\omega/2) \rightarrow 1$, $\vec{r} \rightarrow (0, 0, 1)$ (pure ground state)
- **High temperature** ($T \rightarrow \infty$): $\tanh(\beta\hbar\omega/2) \rightarrow 0$, $\vec{r} \rightarrow (0, 0, 0)$ (maximally mixed)

7 Purity Tests: The Ultimate Quantum Detector

7.1 The Universal Purity Test

The Definitive Purity Criterion

For any density matrix ρ :

$$\text{Tr}(\rho^2) = \begin{cases} 1 & \text{if and only if } \rho \text{ represents a PURE state} \\ < 1 & \text{if and only if } \rho \text{ represents a MIXED state} \end{cases}$$

This test is:

- **Universal:** Works for any quantum system
- **Definitive:** Never gives false results
- **Computable:** Can be calculated from any density matrix
- **Basis-independent:** Result is invariant under unitary transformations

7.2 Mathematical Foundation

Proof of Purity Test

For Pure States: If $\rho = |\psi\rangle\langle\psi|$, then:

$$\rho^2 = (|\psi\rangle\langle\psi|)(|\psi\rangle\langle\psi|) \quad (18)$$

$$= |\psi\rangle\langle\psi|\psi\rangle\langle\psi| \quad (19)$$

$$= |\psi\rangle \cdot 1 \cdot \langle\psi| \quad (\text{since } \langle\psi|\psi\rangle = 1) \quad (20)$$

$$= |\psi\rangle\langle\psi| = \rho \quad (21)$$

Therefore: $\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1$

For Mixed States: Using spectral decomposition $\rho = \sum_i \lambda_i |i\rangle\langle i|$ with $\lambda_i > 0$ and $\sum_i \lambda_i = 1$:

$$\text{Tr}(\rho^2) = \sum_i \lambda_i^2 < \left(\sum_i \lambda_i \right)^2 = 1$$

The inequality is strict when more than one $\lambda_i > 0$.

8 Bloch Vector-Based Purity Tests

8.1 Geometric Purity Formula

Purity-Bloch Vector Relationship

For any qubit density matrix with Bloch vector $\vec{r}$:

$$\text{Tr}(\rho^2) = \frac{1 + |\vec{r}|^2}{2}$$

Proof:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}) \quad (22)$$

$$\rho^2 = \frac{1}{4}(I + \vec{r} \cdot \vec{\sigma})^2 \quad (23)$$

$$= \frac{1}{4}(I + 2\vec{r} \cdot \vec{\sigma} + (\vec{r} \cdot \vec{\sigma})^2) \quad (24)$$

$$= \frac{1}{4}(I + 2\vec{r} \cdot \vec{\sigma} + |\vec{r}|^2 I) \quad (25)$$

$$= \frac{1}{4}((1 + |\vec{r}|^2)I + 2\vec{r} \cdot \vec{\sigma}) \quad (26)$$

Therefore: $\text{Tr}(\rho^2) = \frac{1}{4}(2(1 + |\vec{r}|^2) + 0) = \frac{1 + |\vec{r}|^2}{2}$

8.2 Instant Purity Classification

Bloch Vector Length as Purity Indicator

The length of the Bloch vector instantly determines the purity:

- $|\vec{r}| = 1$: Pure state, $\text{Tr}(\rho^2) = 1$
- $|\vec{r}| = 0$: Maximally mixed state, $\text{Tr}(\rho^2) = \frac{1}{2}$
- $0 < |\vec{r}| < 1$: Mixed state, $\frac{1}{2} < \text{Tr}(\rho^2) < 1$

This provides an immediate geometric test: measure how far the state is from the center of the Bloch sphere!

M. Karthikeya

9 Complete Problem-Solving Protocol

9.1 The 6-Step Quantum State Analysis

Universal Solution Strategy

For any density matrix problem, follow this systematic approach:

Step 1: Validity Check

- Verify $\rho^\dagger = \rho$ (Hermitian)
- Check $\text{Tr}(\rho) = 1$ (normalized)
- Confirm $\rho \geq 0$ (positive eigenvalues)

Step 2: Purity Analysis

- Calculate $\text{Tr}(\rho^2)$
- Classify: Pure ($= 1$) or Mixed (< 1)

Step 3: Pauli Decomposition

- Calculate Bloch vector $\vec{r}$
- Verify $|\vec{r}| \leq 1$
- Write $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$

Step 4: Geometric Representation

- Locate state on Bloch sphere
- Interpret physical meaning

Step 5: Information Analysis

- Calculate Von Neumann entropy if needed
- Assess information content

Step 6: Physical Interpretation

- Connect to physical systems
- Predict measurement outcomes

10 Advanced Applications and Examples

10.1 Entanglement and Partial Traces

Bell State Analysis

System: Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Two-qubit density matrix: $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$ (Pure: $\text{Tr}(\rho_{AB}^2) = 1$)

Reduced density matrix of A:

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \frac{1}{2}|0\rangle_A\langle 0|_A + \frac{1}{2}|1\rangle_A\langle 1|_A = \frac{I}{2}$$

Subsystem purity:

$$\text{Tr}(\rho_A^2) = \frac{1}{2} < 1 \quad \text{MAXIMALLY MIXED!}$$

The Entanglement Paradox:

- **Total system:** Pure state (complete information)
- **Individual subsystems:** Maximally mixed (zero information)
- **Information location:** Stored in quantum correlations between A and B

11 Master Formula Reference

Essential Formulas

Pauli Matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Bloch Representation:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}), \quad |\vec{r}| \leq 1 \quad (27)$$

$$r_x = 2\text{Re}(\rho_{01}), \quad r_y = -2\text{Im}(\rho_{01}), \quad r_z = \rho_{00} - \rho_{11} \quad (28)$$

Purity Tests:

$$\text{Direct test: } \text{Tr}(\rho^2) = 1 \text{ (pure)}, < 1 \text{ (mixed)} \quad (29)$$

$$\text{Bloch test: } \text{Tr}(\rho^2) = \frac{1 + |\vec{r}|^2}{2} \quad (30)$$

$$\text{Geometric test: } |\vec{r}| = 1 \text{ (pure)}, < 1 \text{ (mixed)} \quad (31)$$

Thermal States:

$$\rho_{\text{thermal}} = \frac{1}{Z} e^{-\beta H} \quad (32)$$

$$\vec{r}_{\text{thermal}} = (0, 0, \tanh(\beta \hbar \omega / 2)) \quad (33)$$

12 Conclusion and Physical Insights

This comprehensive framework provides a complete mathematical and geometric understanding of quantum states through the unified lens of Pauli matrices, Bloch sphere visualization, and purity analysis. The key insights include:

1. The **universal purity test** $\text{Tr}(\rho^2)$ definitively distinguishes pure from mixed states
2. The **Pauli decomposition** $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$ connects abstract matrices to geometric visualization
3. The **Bloch sphere** provides intuitive geometric representation of all single-qubit quantum states
4. **Bloch vector length** instantly reveals state purity: $|\vec{r}| = 1$ (pure), $|\vec{r}| < 1$ (mixed)
5. **Mixed states** arise naturally from thermal equilibrium, decoherence, measurement, and entanglement
6. The framework seamlessly connects microscopic quantum mechanics to macroscopic physical systems

This unified approach forms the foundation for understanding quantum computing, quantum information theory, quantum thermodynamics, and modern quantum technologies. The elegant connection between abstract mathematical formalism and geometric intuition through the Bloch sphere makes quantum mechanics more accessible while maintaining complete mathematical rigor.

Machiraju Karthikeya
Advanced Quantum Mechanics & Quantum Information

"The future belongs to those who understand the quantum realm"

M. Karthikeya